

Samyu Manchikanti

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PROFESSIONAL SUMMARY

- Data Scientist with 4 + Years of experience executing data-driven solutions with adept knowledge on Data Analytics, Data Modelling, Text Mining, Machine Learning (ML), Predictive Modelling, and Natural Language Processing (NLP).
- Experience with Python Libraries Pandas, NumPy, Seaborn, NLTK, and Scikit - learn Keras, and TensorFlow in developing end-to-end Analytics and ML, DNN models.
- Knowledge on Natural Language Processing (NLP) and Multi-layer perceptron algorithm and Text Mining.
- Worked and extracted data from various database sources like Oracle, PostgreSQL, MySQL, and SQL Server.
- Proficient in Machine Learning & Deep Learning algorithms, and Predictive Modelling including Regression Models, Decision Trees, Random Forests, AI-Learning, Transfer Learning.
- Have hands-on experience in Spark SQL and usage of Hadoop Architecture frameworks and various components.
- Experience with deploying models on the Azure Databricks cloud.
- Actively engaged in the entire data science project life cycle, from data extraction and data cleaning to statistical modelling and data visualization, handling extensive sets of both structured and unstructured data.
- Strong skills in statistical methodologies such as A/B test, experiment design, hypothesis test, ANOVA.
- Skilled in Advanced Regression Modelling, Correlation, Multivariate Analysis, Model Building, Business Intelligence tools and application of Statistical Concepts.
- Experience in implementing data analysis with various analytic tools, such as Anaconda 4.0, Jupyter Notebook 4.X, R 3.0 (ggplot2, Caret, matplotlib), Spyder, Pycharm.
- Proficient in data visualization tools such as Tableau, and Power BI to create visually powerful and actionable interactive reports and dashboards.
- Experienced in Big Data with Hadoop, HDFS, MapReduce, and Spark.
- Expertise in writing complex queries on SQL, Postgres, and Oracle.
- Experienced in Data Integration Validation and Data Quality controls for ETL process.
- Strong skills in Linear algebra, Probability theory, and Calculus.
- Implemented Bagging and Boosting to enhance the model performance.
- Experienced the full software life cycle in SDLC, Agile, DevOps and Scrum methodologies including creating requirements, test plans.
- Strong business sense and ability to communicate data insights to both technical and nontechnical clients.
- Excellent interpersonal and communication skills. Good documentation and process management skills with the ability to effectively understand the business requirement to develop a quality product.

WORK EXPERIENCE

Sr. Data Scientist/ML/AI Engineer Clarkston Consulting

Aug2023 to April 2024
Durham, NC

- Explored and analysed the customer specific features by using Matplotlib, Seaborn in Python and dashboards in Tableau.
- Built a data warehouse by utilizing ETL processes with tools such as Apache Nifi and Apache Hadoop to gather all the business data related to doctors, patients, prescriptions, orders and calls from different sources.
- Developed doctor report cards for real-time insights into their performance over the years. Using Apache Kafka for data ingestion and Tableau integrated with Hadoop/Spark for creating the reports.
- Developed predictive models like disease risk, readmission risk using advanced machine learning algorithms, ensemble models, and deep learning architectures.
- Used Pandas, NumPy, Scikit-learn in Python for developing various like Emergency department wait time, chronic disease progression machine learning models and utilized algorithms such as Linear regression, Logistic regression, Gradient Boosting, SVM and KNN.
- Developed pipeline using Hive (HQL) to retrieve the data from Hadoop cluster, SQL queries to retrieve data from MySQL database and used ETL for data transformation.
- Derived data from relational databases to perform complex data manipulations and conducted extensive data checks to ensure data quality. Performed Data wrangling to clean, transform and reshape the data utilizing NumPy and Pandas library.
- Implemented installation and configuration of multi-node cluster on Cloud using AWS.
- Utilized IOT sensors for collecting health information of cold storage and build streaming data pipeline into GCP's BigQuery with the help of Apache Airflow.
- Productionized machine learning pipelines to gather data from BigQuery using Apache Airflow and build forecasting models to predict temperature and humidity spikes inside the cold storage.
- Built monitoring dash boards by employing visualization tools such as Tableau or Power BI, visualizing both the current state and predictive health of cold storage warehouses.
- Applied NLP techniques for sentiment analysis on customer feedback and reviews.
- Implemented NLP algorithms to analyse customer interactions and provide personalized health and wellness recommendations.
- Utilized NLP for understanding and extracting information from clinical documents and medical records.
- Led the development and deployment of machine learning models on GCP Vertex AI tailored for healthcare applications, including

predictive analytics for patient outcomes and disease progression.

- Utilized Vertex AI's AutoML capabilities to fine-tune models for medical image analysis, ensuring high accuracy in tasks such as radiology image interpretation.
- Designed end-to-end machine learning pipelines on GCP Vertex AI with a focus on security and compliance, ensuring that healthcare data handling adheres to regulatory standards like HIPAA.
- Implemented encryption and access controls within Vertex AI to safeguard patient data during model training, evaluation, and inference stages.
- Led the development of Deep Learning models, utilizing PyTorch and Tensor flow, to address intricate challenges and enhance predictive capabilities.
- Leveraged Python, PyTorch, and Tensor flow to design and implement cutting edge models, enhancing the organization's capabilities in applied research and data-driven decision-making.
- Utilized Azure tech stack for secure storage and management of Electronic Health Records (EHRs).
- Applied OpenAI's natural language processing capabilities to analyzed and understand unstructured clinical notes.
- Implemented Hugging Face's Transformers library to enhance natural language understanding in customer interactions.
- Utilized pre-trained language models to improve chatbot responses and customer support services.
- Implemented the application of retrieval augmented generation techniques to optimize customer support chatbots. Implemented advanced methods to retrieve historical customer interactions and generate context-aware responses, significantly improving the accuracy and responsiveness of the chatbot in addressing user queries.

Environment: Python, R, HDFS, ODS, OLTP, Power BI, Tableau, Hive, AWS, Big Query, Apache Airflow, Hadoop, MySQL, OLAP, DB2, Metadata, Tera Data, MS Excel, Mainframes MS Vision, Spark, Map-Reduce, SQL and MongoDB

Data Scientist

Aug 2020 to Oct 2022

Blue5green

India

- Identify business problems or management objectives that can be addressed through data analysis and propose creative solutions and strategies to existing business challenges.
- Analyze, manipulate, and process massive amounts of data using statistical software to discover trends, patterns, and insights via Jupyter, Sci-kit learn and Tableau.
- Automate the entire collection process pipeline by identifying valuable data sources by using ETL tools like Apache Nifi, Apache Beam.
- Apply feature selection algorithms to models such as ANOVA (analysis of variance), decision trees using PySpark's Mllib package and hyper tune the parameters based on interest and to predict the outcomes.
- Design and implement predictive learning models and machine learning algorithms like K-NN, Logistic regression etc., for financial applications like- Transaction Classification and Risk Modelling and combine them through ensemble learning.
- Present data using data visualization techniques by using tools like Tableau and libraries like matplotlib, ggplot2, seaborn by creating graphs, charts, or other visualizations to convey the Results of data analysis.
- Read research and scientific articles, conference papers, or other sources to identify emerging analytic trends and technologies to improve efficiency of existing models.
- Perform Quality Analysis testing and validation internally using Django and reformulate models to ensure accurate prediction of outcomes of interest and end to end API testing with dummy data and actual data before launch of the actual product with the Engineering team.
- Prepare documentations and deliver oral or written presentations of the results of mathematical modelling and data Analysis to stakeholders.
- Applied mean-variance optimization algorithms, such as Markowitz portfolio theory, using optimization libraries like scipy. Optimize in Python, to construct efficient investment portfolios balancing risk and return.
- Implemented anomaly detection algorithms, such as isolation forests and autoencoders, with Python libraries like scikit-learn and TensorFlow/Keras to detect and prevent fraudulent activities in financial transactions.
- Utilized natural language processing (NLP) techniques, including sentiment analysis and named entity recognition, with NLP libraries like NLTK and spaCy in Python, to analyze regulatory texts and ensure compliance.
- Employed clustering algorithms, such as k-means and hierarchical clustering, using Python libraries like scikit-learn, to segment customers based on demographic and behavioural data for targeted marketing campaigns.
- Developed credit risk models using gradient boosting machine algorithms, such as XGBoost and LightGBM, in Python, to assess creditworthiness and predict default probabilities for loan applicants.
- Employed model evaluation techniques, including cross-validation and performance metrics like ROC-AUC and F1 score, using Python libraries like scikit-learn, to continuously improve predictive models based on feedback and data updates.
- Utilized version control systems like Git and collaboration platforms like Jira to facilitate seamless collaboration with cross-functional teams and track project progress efficiently.

Environment: Python, PySpark, Django, Tableau, PySpark, Apache Nifi, Apache Beam, ML algorithms, Jupyter.

TECHNICAL SKILLS

- **Languages:** Python, R, C, C++, SQL, R, MATLAB, SAS PostgreSQL, Oracle, MySQL, Cassandra, SQL Databases
- **Framework & Libraries:** TensorFlow, Keras, NumPy, Scikit-Learn, Pandas, Seaborn, OpenCV, PyTorch, Pyspark, KNN, K-Medoids, NLP, SVM, K-Means, MLP, Random Forest, Databricks, Matplotlib ggplot
- **Other Skills:** PCA, Neural Networks, Deep Learning, Supervised Learning, Unsupervised Learning, Predictive Analysis, Reinforcement Learning, Transfer Learning.
- **Visualization Tools:** Power BI, Tableau, Qlik
- **Tools:** Hadoop, Spark, KNIME, Kafka, SPSS, ETL (Informatica, Pentaho, Knime), MongoDB, Docker
- **Cloud:** AWS, Amazon Sagemaker, Azure, Databricks, GCP
- **Operating Systems:** Windows, macOS, Linux
- **Platform:** GIT, Bamboo, Stash, Jira, Agile methodology (Scrum)

EDUCATION

Master of Science in Information Technology from University of North Carolina at Charlotte **Jan2023 – May 2024**
Bachelor's in Information Technology from TKR College of Engineering and Technology **Aug 2017 - May 2021**